

Research Vision

Mathematics without biology or biology without mathematics are not as wonderful.

My *NeuroAI Research Group* will initially focus on creating advanced statistical methods to characterise neural and behavioral data, offline and in real time, for short- and long-duration experiments. In addition, in collaboration with experimental groups, we will apply these methods, or those created by others, to investigate behavioural and neural questions.

This is an exciting time to work on neuroAI. As the variety of neural datasets being recorded (e.g., high-channel electrophysiology, one- and two-photon calcium imaging, voltage imaging, EM, transcriptomics, connectomics) keeps increasing, and their size and complexity become larger and larger due to more naturalistic and longer experimentation, there is a critical need for statistical methods that can draw efficient inferences and extract meaning from these datasets. In addition, as new optogenetic methods are developed to spatio-temporally manipulate neural activity, we need these inferences in real time, combined with methods from optimal control, to maximise the efficiency of these manipulations. Yet, we are currently not seeing an increase in the number of scientists trained in both statistical machine learning and neurosciences required for the development of these methods. My research group will attempt to reverse this trend.

Latent variable models It is challenging to extract meaning from the large number of spikes generated by high-channel count recordings probes, like Neuropixels. To address this problem, my supervisor at the Gatsby Unit, Prof. Maneesh Sahani, created Gaussian Process Factor Analysis (GPFA; [Yu et al., 2009](#)), a probabilistic method to extract low-dimensional representations (i.e., latent variables) from high-dimensional electrophysiological recordings, and recently extended it in Sparse Variational Gaussian Process Factor Analysis (svGPFA; [Duncker and Sahani, 2018](#)). At the Unit I have distributed a highly optimized Python implementation of svGPFA and explored its capabilities to [characterize several high-channel count neural datasets](#).

In collaboration with the laboratory of Marcus Stephenson-Jones at the SWC, I have extracted svGPFA neural latents from striatal Neuropixels recordings in mice performing a stereotyped poking sequence task ([Thompson et al., 2024](#)). I then used these neural latents as inputs to a Hidden Markov Model to decode mice behavioral states with very high accuracy. We are now observing that patterns of neural latents detected during behavior replay during sleep. Similar decoding and replay are obtained using latent variables estimated using linear dynamical systems, instead of svGPFA. If these observations are confirmed, it will be the first time that accurate decoding is achieved and replay is detected in continuous latent variables of large neural populations, bypassing the estimation of place fields.

In collaboration with Prof. Maneesh Sahani, my group will continue developing svGPFA and will collaborate with experimental groups at NPP and CDB to explore the potential of latent variable models for position decoding and replay detection in the hippocampus.

NeuroAI methods for naturalistic, long-duration and continual experimentation At the SWC and Gatsby Unit, we are pioneering a new type of naturalistic, long-duration and continual experimentation, the Aeon platform, that will revolutionise neuroscience experimentation ([Campagner et al., 2025](#)).

In 2023-2024 I mentored a very talented undergraduate student from NPP, [Ms. Zimo Li](#), in a [Simons Foundation SURFiN fellowship](#), where she performed the first comprehensive statistical analysis of Aeon data. The achievements of Zimo were outstanding. She processed very complex data, that at that stage were poorly organised, and skillfully applied sophisticated statistical models (e.g., linear dynamical systems, hidden Markov models, point process regression) and produced the first statistical Aeon discoveries, which were included in the [Aeon platform paper](#), with Zimo as coauthor. The collaboration with Zimo taught me that I can do excellent research with students that do not have a solid mathematical background.

In collaboration with the Aeon team at the Gatsby Unit and SWC, my group will continue developing statistical methods for the analysis of Aeon data. Due to the large size of Aeon recordings, conventional batch machine learning (ML) methods cannot be used to process Aeon data. Aeon requires online methods that are able to process non-stationary data and are computationally very efficient to handle very large datasets. A detailed description of my ML data processing plans for Aeon data appears in a [collaborative BBSRC/NSF grant we submitted jointly with the Allen Institute for Neural Dynamics in 2024](#).

NeuroAI methods for intelligent experimental control Current neuroscience experimentation is mostly performed in open loop with predefined sets of rules designed by experimenters. More informative experiments can be designed if they could be controlled in close loop by intelligent inferences performed on behavioural and neural data streams.

[Bonsai](#) is today the most widely used software for close loop experimental control in neuroscience. In 2022 I realised that adding ML functionality to Bonsai could make possible a radically new type of close loop intelligent experimentation, and led the successful application of a BBSRC proposal ([BB/W019132/1](#)) to build the [Bonsai.ML package](#).

Most current ML methods operate offline, trained and performing inference on datasets previously recorded. However, Bonsai.ML requires a different type of ML methods, which are trained and perform inference on live streams, continuously

delivering data in real time. We built this type of ML into Bonsai.ML. It now contains probabilistic ML methods for real time processing of behavioral (e.g., kinematics inference with linear dynamical systems, behavioral state estimation with hidden Markov models) and neural data (neural latent variables inference with linear dynamical systems, hippocampal position decoding and replay detection with a point process state-space model). Please refer to the [documentation of Bonsai.ML](#).

We are now integrating large language models (LLMs) into Bonsai. We are using them to help Bonsai users build correct and optimised workflows using natural language. Additionally, we are building a Bonsai LLM node to bring the large semantic knowledge of LLMs into the experimental controls loop. Please refer to the [Open Source for the Life Science proposal](#) we are currently preparing to support these developments.

My group will continue developing real-time probabilistic ML models to enable intelligent experimental control in Bonsai.

Collaborations In 2019 the Gatsby Unit hired me to facilitate collaborations between the Unit and the SWC. At the SWC experimental scientists collect rich and complex neural and behavioral datasets. Because often they don't have the analytical skills to best characterise these recordings, they end up using simple and sub optimal statistical techniques. I greatly enjoyed collaborating with SWC experimental scientists finding together good statistical tools to characterize their data. Below I highlight one of these collaborations.

To explore collaborative opportunities I attended the SWC Data Clubs, where SWC scientists presented their latest findings. In one data club Ms. Mitra Javadzadeh, Ph.D. student in Prof. Hofer's lab, presented her long-range cortical communications research, where she was using Linear Discriminant Analysis to model this communication. I proposed to use linear dynamical systems instead, as the dynamic characteristic of this model better suited her experimental setup. Mitra shared with me a sample of her data and asked me to try my suggestion on it. COVID time was ideal for this analysis, which produced very interesting results. I shared these initial results with her, her advisor and my supervisor joined the investigation, and [Mitra presented orally our results at CoSyNe 2022¹](#). Now, as faculty member at CSHL, Mitra continues this line of research ([Javadzadeh et al., 2024](#)).

I find the research of NPP and CDB very interesting and look forward to collaborate with their investigators on the use of statistical models to characterize their data. Specifically, at NPP I see opportunities for collaborations with the lab of Prof. Bianco on methods to process calcium imaging data; with Prof. Bush on methods to process hippocampal data (e.g., decoding, forecasting and replay detection), on EEG data processing, and on methods for seizure detection and prediction; with Prof. MacAskill on RL methods supporting context-dependent decision making and methods to process calcium imaging data; with Prof. Ruediger on methods to characterize response properties of visual cells. At CDB I see opportunities to collaborate with the groups of Prof. Barry and Prof. O'Keefe on methods to process hippocampal data; with Prof. Baier on dynamical systems and epilepsy; with Prof. Cohen on methods for real-time experimental control.

Teaching Vision

I am experienced teaching quantitative subjects to life sciences students. Please refer to the Teaching section of my CV.

My teaching philosophy has been heavily influenced by my postdoctoral mentor, Prof. Wilson Truccolo, Brown University, and aligns closely with the style of [Case Studies in Neural Data Analysis](#), by Prof. Mark Kramer, Boston University. I deliver lectures that are mathematically rigorous, but skip proofs from the main thread of content. These proofs are provided in appendices. I complement my lectures with many examples, most often in the form of computer code. These examples help students grasp the utility of the concepts I teach, even if they don't fully understand their mathematical underpinning. I design my lectures to be multidimensional, so that both naive and expert students can benefit from them. Please refer to the [Foundations of probability theory](#) lecture, delivered in the first edition of Gatsby Unit's Bridging Program in 2023, as a demonstration of my teaching style.

I am very enthusiastic about learning, and from students' feedback I learned that some of this enthusiasm transpires in my teaching.

¹Mitra's presentation starts at 2:59:00

References

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